Title: Wi-Fi network scanning using mobile apps (SSID, RSSI, channel mapping).(WiFi Analyzer)

Theory: Wi-Fi network scanning is used to identify nearby wireless networks and analyze their characteristics.
The important parameters are:

- SSID: Name of the wireless network.
- RSSI: Received Signal Strength Indicator, measured in dBm.
- Channel: Frequency channel used by the Wi-Fi router.
- Frequency band: Usually 2.4 GHz or 5 GHz.

Observations :

Sr. No.	SSID	BSSID	RSSI)dBm)	Channel	Frequency (GHz)	Security
1	FAMT-WiFi	54:b8:0a:3c:45:d8	-54	1	2,412	Open
2	iQOO Z9 5G	3e:94:d4:2d:67:ca	-64	1	2,412	WPA2
3	Redmi Note 13 5G	aa:3e:3c:1f:98:20	-64	11	2,462	WPA2
4	FAMT-WiFi	c4:a8:1d:93:7d:98	-93	11	2,462	Open
5	Tazmeen	46:a3:b2:72:2f:0f	-92	6	2,437	WPA2

conclusion: Wi-Fi network scanning was performed using a mobile application. Nearby wireless networks and their signal strengths, channels, and frequency bands were observed.

Title: Measurement of signal strength vs distance for Wi-Fi router.(WiFi Analyzer)
Theory: Wi-Fi is a wireless communication technology used to connect devices to a network using radio waves.
The strength of a Wi-Fi signal is measured using RSSI (Received Signal Strength Indicator), expressed in dBm.
As the distance between the Wi-Fi router and the receiving device increases, the signal strength generally decreases due to signal attenuation.Signal strength is also affected by:

- Distance between the router and receiver.
- Walls and other physical obstacles.
- Interference from nearby wireless devices.
- Router antenna and environmental conditions.

Observations:

Table 1: RSSI vs Distance

Distance (Steps)	Approximate Distance (m)	RSSI (dBm)	Signal Quality (Excellent/Good/Fair/Poor)
0 steps	0 m	-35 dBm	Excellent
5 steps	~3.75 m	-45 dBm	Excellent
10 steps	~7.5 m	-53 dBm	Good
15 steps	~11.25 m	-61 dBm	Good
20 steps	~15 m	-68 dBm	Fair
25 steps	~18.75 m	-74 dBm	Fair
30 steps	~22.5 m	-80 dBm	Poor

Conclusion:
The experiment shows that Wi-Fi signal strength decreases as the distance between the router and smartphone increases. The RSSI changed from a strong value near the router to a weaker value at greater distances

Title: Throughput and latency measurement using network tools
Theory: Throughput is the amount of data successfully transmitted over a network per unit time. It is usually measured in Mbps.
Latency is the time taken for a data packet to travel from the source to the destination and back. It is measured in milliseconds (ms) using tools such as `ping`.

Observation Table 1: Ping Test

Sr. No.	Destination	Packet Size (bytes)	Time/Latency (ms)	TTL	Status
1	google.com	32	18	117	Reply received
2	google.com	32	20	117	Reply received
3	google.com	32	19	117	Reply received
4	google.com	32	21	117	Reply received

Observation Table 2: Speed Test

Test No	Download Latency (ms)	Upload Latency (ms)	Download Speed (Mbps)	Upload Speed (Mbps)
1	20	25	48.5	18.2
2	18	28	51.2	19.5
3	21	27	49.8	18.9
Average	19.7	27	49.8	18.9

Conclusion: Network throughput and latency were measured using network tools. Throughput indicates the data transfer performance, while latency indicates the responsiveness of the network.